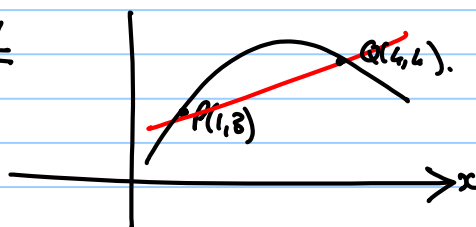


Unit 2.1 Tangent and Velocity Problems. (p. 78-82)

Example 1

Find the equation of the secant line PQ:

$$y = mx + c.$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 3}{4 - 1} = \frac{1}{3}$$

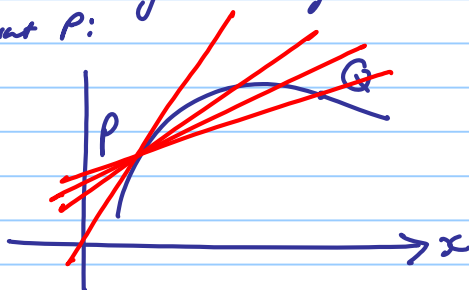
$P(1, 3)$
 $Q(4, 4)$

$$\Rightarrow y = \frac{1}{3}x + c$$

$$4 = \frac{4}{3} + c \Rightarrow c = 4 - \frac{4}{3} = \frac{12-4}{3} = \frac{8}{3}$$

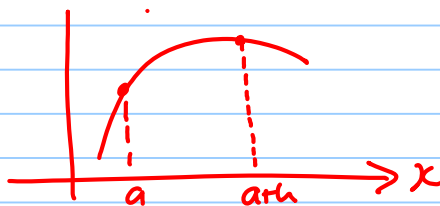
$$\Rightarrow y = \frac{1}{3}x + \frac{8}{3}$$

Moving Q now closer and closer to P results in PQ becoming the tangent line to the curve in the point P:



The slope of the tangent of P is equal to the instantaneous rate of change.

If we have:



Average rate of change: $\frac{f(a+h) - f(a)}{h}$

Instantaneous rate of change: $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

$\therefore$ we need to be able to find limits.
The following section is on calculating limits.

Moving Q closer and closer to P , results in PQ becoming the tangent line to the curve in the point P .

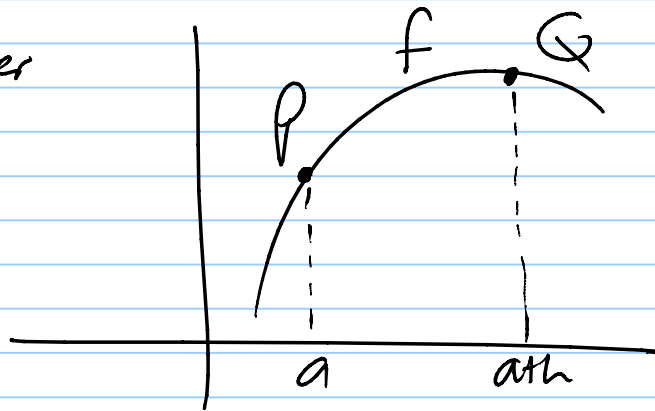
Slope of secant PQ
= average rate of change

=

Slope of tangent at P
= instantaneous rate of change

[2]

Consider



Avg rate of change =

Instantaneous rate of change

=